Order Flow Imbalance

A microstructure study of OFI as a price-predictive signal

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Summary

Order Flow Imbalance (OFI) measures net buying versus selling pressure at the top of the limit order book. Using 300,000 simulated book events (8.3 hours, roughly a one-tick spread), this note finds that OFI explains a large share of contemporaneous one-second price moves and retains some forecasting power out of sample, but the predictable move is smaller than the spread, so a liquidity taker cannot trade it profitably after costs. A market maker can use the same signal to skew quotes and reduce adverse selection.

- OFI explains 53% of contemporaneous one-second price moves ($R^2 = 0.53$, $t = 132$), or 88% using the top five levels.
- It forecasts the next second weakly (IC = 0.18) and holds out of sample (IC = 0.19).
- The predicted move (0.09 ticks) is about 29 times smaller than the round-trip taker cost (2.48 ticks), so a cost-aware taker makes 0 trades.
- Skewing Avellaneda-Stoikov quotes by OFI keeps inventory tightly bounded (± 15 versus ± 173 for naive quoting) while recovering most of the PnL that plain inventory control gives up ($110 versus $18), the best risk-adjusted return of the three policies.

1 Price impact

A regression of the per-second mid-price change on OFI (Cont, Kukanov and Stoikov, 2014) gives $R^2 = 0.53$ ($t = 132$); the near-linear relationship is consistent with Kyle-type linear price impact (Kyle, 1985). Using OFI from the top five book levels (Xu, Gould and Howison, 2020) raises this to $R^2 = 0.88$.

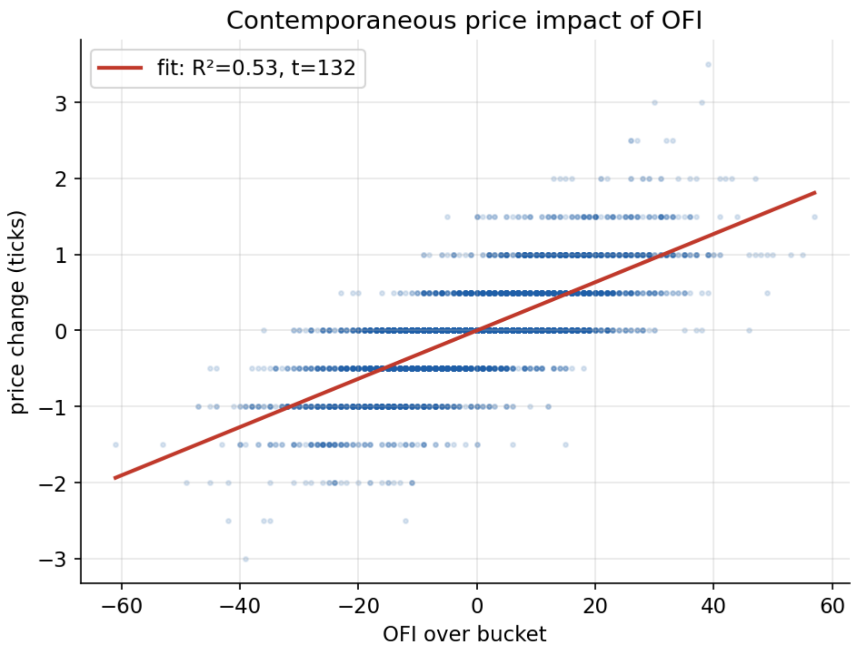

Figure 1: Per-second price change versus OFI.

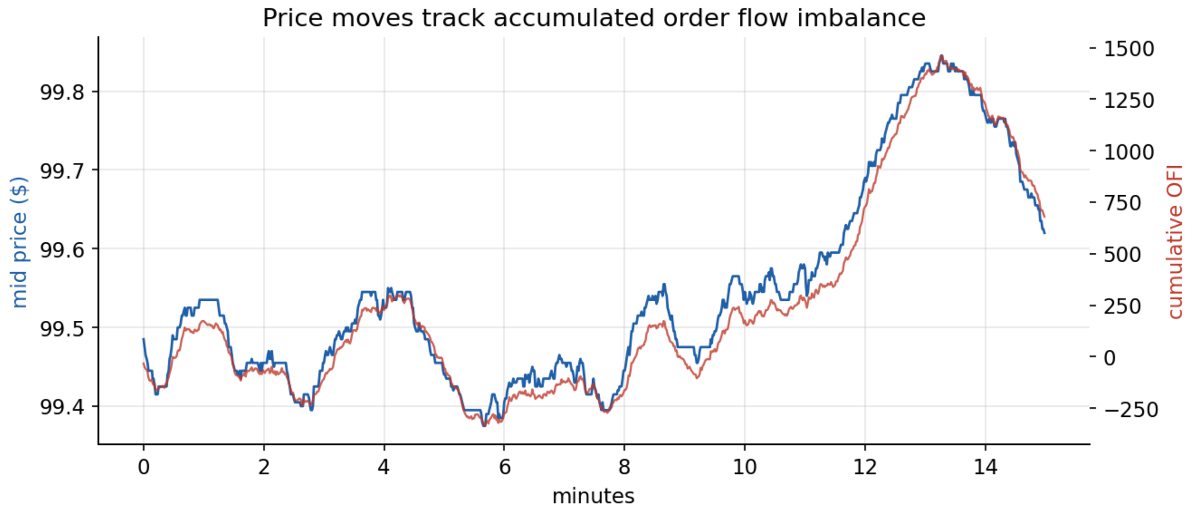

Figure 2: Mid price and cumulative OFI over a 15-minute window.

2 Forecasting

Forecasting the next move is weaker. The one-second information coefficient is 0.18 and holds out of sample (in-sample 0.18, out-of-sample 0.19 on a 70/30 chronological split). The IC rises with horizon, which indicates OFI predicts a persistent drift rather than a single move.

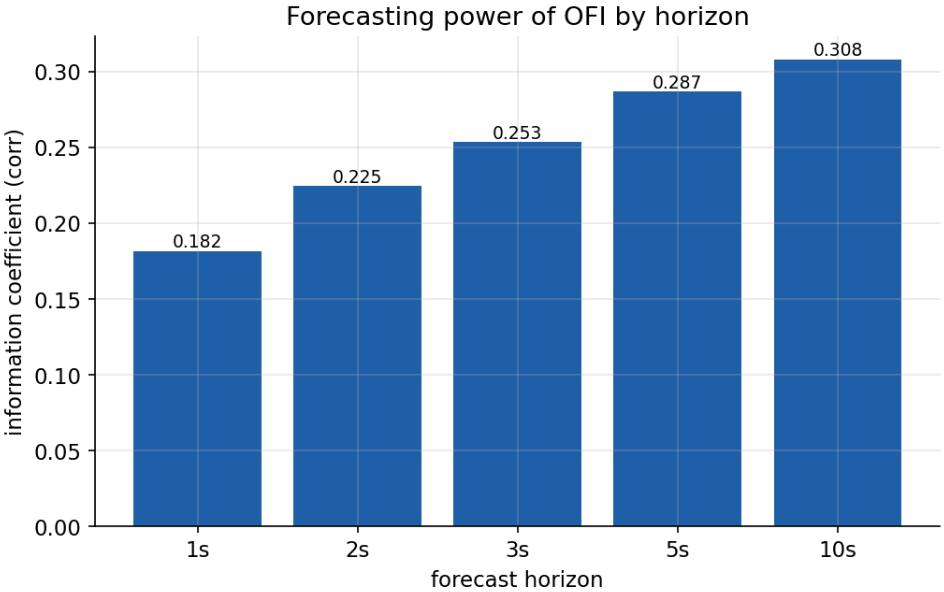

Figure 3: Information coefficient by forecast horizon.

3 Transaction costs

The predicted next-second move averages 0.09 ticks. A taker round trip costs about 2.48 ticks (two spread crossings plus fees), roughly 29 times the edge. Out of sample the signal is positive gross (+0.29 ticks per trade) but negative net (−2.20 ticks per trade); a taker that trades only when the predicted edge exceeds the cost makes 0 trades.

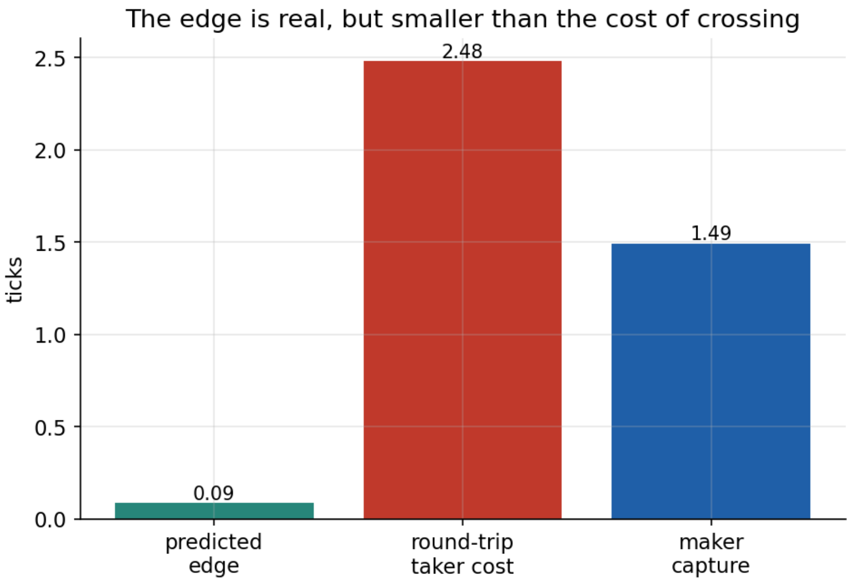

Figure 4: Predicted edge, taker cost, and maker capture, in ticks.

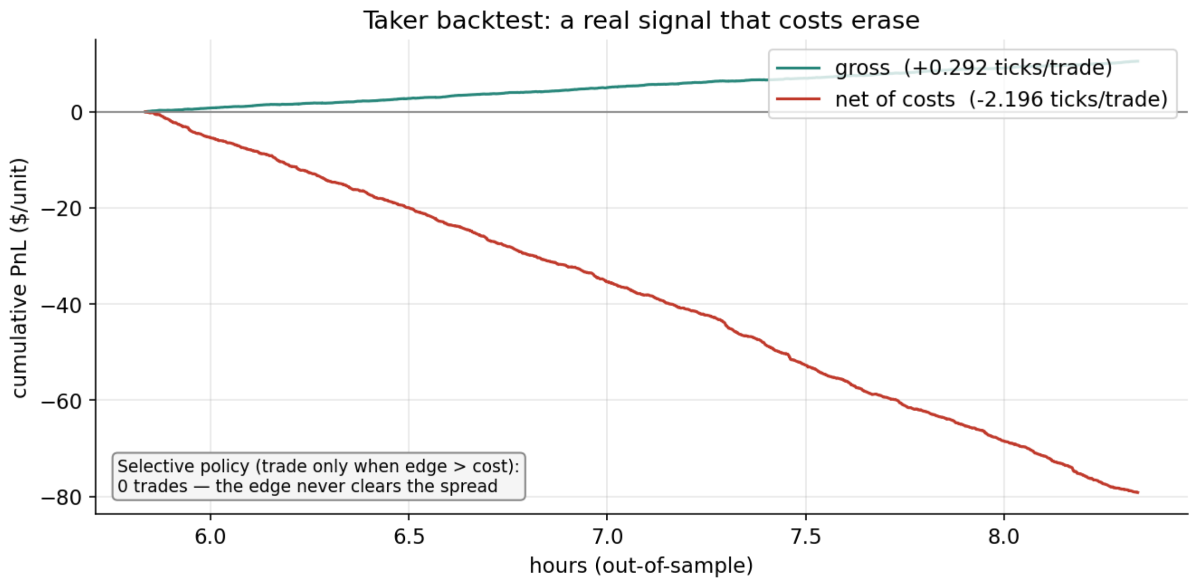

Figure 5: Out-of-sample taker backtest, gross and net of costs.

4 Market making

A market maker earns the spread but must control inventory and avoid being run over by informed flow (Glosten and Milgrom, 1985). Plain Avellaneda-Stoikov quoting (Avellaneda and Stoikov, 2008) bounds inventory but gives up most of the PnL; adding an OFI skew to the reservation price uses the forecast to position inventory ahead of the move and recovers the PnL at the same low inventory. Three policies are compared on the same price path.

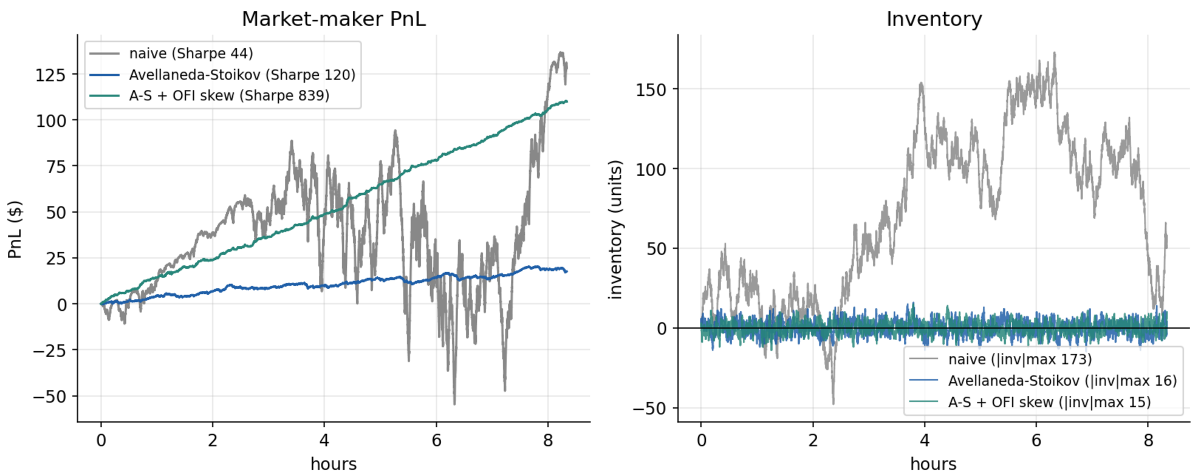

Figure 6: PnL (left) and inventory (right) for the three quoting policies.

Policy	Final PnL	Max abs. inv.	Sharpe*
Naive symmetric	$128	173	44
Avellaneda-Stoikov	$18	16	120
A-S + OFI skew	$110	15	839

*Sharpe is annualised from one-second marks and is large for all policies; the meaningful comparison is across policies, not the absolute level.

5 Method and limitations

Forecasting labels use the price move after OFI is observed, costs are charged at the spread prevailing at trade time, and the model is fit on the training period and evaluated on the test period. Standard errors use the Newey-West HAC estimator (Newey and West, 1987) because high-frequency returns are autocorrelated.

The order book is a calibrated simulator; a live Binance collector is included so the same pipeline can run on real data. The fill model is an Avellaneda-Stoikov intensity rather than a queue-position simulation, and second-level Sharpe figures are not comparable to fund-level Sharpes.

References

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